

1(a)(i)	translation ;	1
1(a)(ii)	A – messenger RNA / mRNA ; B – transfer RNA / tRNA ; C – (poly)peptide (chain) ; I PPC / (primary structure of) protein / amino acid chain	3
1(a)(iii)	<i>accept names of bases</i> D – AAA E – GUG ;	1

Question	Answer	Marks
1(b)	any four from: <i>structure to max three</i> <i>ignore all refs to DNA and red blood cells</i> - different amino acid will have different, R group / side chain ; A glutamic acid / glu, to, valine / val - may prevent formation of bond(s) between R groups ; - may form, new / different, bond(s) between R groups ; - change in, secondary / 2° / tertiary / 3° / quaternary / 4°, structure ; A change in, globular shape / 3D shape A different folding of, β-globin / polypeptide / haemoglobin R quaternary if referring to, a single polypeptide / β-globin - any further detail on, 2° / 3° / 4°, structure ; e.g. interactions between α-globin and β-globin may change / AW - AVP ; e.g. change from, polar / charged / hydrophilic, to, non-polar / non-charged / hydrophobic (R group / amino acid) <i>function</i> - change in ability, to bind / AW, to oxygen / carbon dioxide ; A lower / higher for change A change in percentage saturation of haemoglobin with oxygen A change in affinity for, oxygen / carbon dioxide A changes the volume of, oxygen / carbon dioxide, carried A carry, more / less, oxygen / carbon dioxide R if haemoglobin is said to have an active site	4